Highlights

Multi-Agent LLM Decision Support for Vietnamese Equities: A Historical Evaluation[⋆]

- Frozen calls support an auditable historical LLM evaluation.
- Class-aware baselines contextualize results on 102 market units.
- Repeated executions expose variability and ordering limitations.
- Negative account returns limit economic and deployment claims.

Multi-Agent LLM Decision Support for Vietnamese Equities: A Historical Evaluation

Abstract

We evaluate a multi-agent LLM implementation for Vietnamese equity decision support using frozen historical calls and class-aware numerical controls. The study covers 102 market units across nine equities, twelve selected dates and six blocks, with 25 UP and 77 DOWN net-return labels. Its Full–Base comparison changes both factor and sentiment sections and is treated as a bundle contrast. Full balanced accuracy is 52.96%, compared with 49.71% for Base and 55.61% for a price-only logistic comparator. CASH has ordinary accuracy 75.49% but balanced accuracy 50%. Mean separate-account net returns are -0.797% for Full and -0.750% for Base. Scope was amended after partial inference, making the study exploratory. Two distinct nine-unit panels retain three executions and a past-donor factorial placebo, with small support and ordering/epoch confounds. An auxiliary train-fit numeric study does not repair the native selector's row-wise causal defect. Financial-news domain validity, historical news availability and immutable hosted-model revisions remain unverified. We archive inputs, requests and outcomes for saved-prediction replay, while public distribution rights and a persistent identifier remain pending. The evidence supports a bounded historical implementation assessment, not causal alpha attribution, general predictive superiority or deployable profitability.

Keywords: Multi-agent LLM, Vietnamese equities, Historical evaluation, Class imbalance, Reproducibility

1. Introduction

Large language models can turn numerical indicators, news and chart images into financial narratives, but a coherent narrative is not necessarily an informative forecast. Evaluation must distinguish a change in class preference from improved discrimination and must distinguish simulated account outcomes from executable trading returns. These distinctions are especially important when transaction costs make the profitable-LONG class rare and when several equities share the same decision dates.

We examine a concrete multi-agent implementation for Vietnamese equities. It combines deterministic technical-indicator reports, a native selector over 85 OHLCV factor candidates, Vietnamese sentiment scores, two chart-reading roles and a final binary Decision Agent. Indicator, Pattern, Trend and Decision roles and associated utilities reuse or modify QuantAgent components [1]. The integration is an implementation and domain-adaptation exercise: condensed reports enter a prompt with heuristic evidence-priority instructions, rather than a learned knowledge-fusion representation.

Our question is how this implementation compares with class-aware and numerical controls on the same recorded historical market units, and what its archived executions can support about variability, components and economics. The primary comparison is Full versus Base. Base removes

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both selected-factor and sentiment sections, so this is a bundle contrast, not an isolated alpha-information treatment. A separate factorial panel includes factor-only, sentiment-only and matched-structure past-donor variants. Different prompt families and small repeated panels remain explicit rather than being pooled into a larger apparent experiment.

The completed release contains 102 main market units spanning six historical blocks and twelve selected dates, with 25 UP and 77 DOWN net labels. Full has pooled balanced accuracy (BA) of 52.96%, Base 49.71%, and a price-only logistic comparator 55.61%. CASH has higher ordinary accuracy than Full but BA of 50%. Mean reference account returns are negative for Full and Base. These observations motivate an assessment of the implementation and its limits, rather than a claim of profitable trading or general predictive superiority.

The evaluation also exposes unresolved methodological defects. Native factor operations are isolated from prices after the outer decision cutoff in synthetic probes, but are not causal row by row within their trailing history. Native ranking and orientation diagnostics are in-sample. News availability and the sentiment classifier's financial-domain validity are not independently authenticated. We preserve these facts when reporting the frozen outputs. Separate train-fit numeric controls use causal adapters and cannot retrospectively repair the native LLM study.

The contributions are therefore an auditable implementation description, a common-support comparison including simple controls, and an explicit accounting of repeat, provenance and execution limitations. Historical snapshots, actual requests/responses and metrics are retained with hashes for offline replay. The evidence is historical and exploratory: the reduced scope was amended after some inference, all outcome dates preceded the freeze, and the study does not establish causal component value or prospective deployment performance.

2. Related work

Financial LLMs and chart agents. Financial language models and adaptation frameworks include BloombergGPT and FinGPT [2, 3]. TradingAgents organizes multiple financial analysis and decision roles [4]. QuantAgent's version-specific architecture already includes Indicator, Pattern, Trend and Decision agents, shared state, chart rendering and trendline utilities [1]. Our system reuses or modifies those elements; adding an alpha/sentiment path does not make the underlying chart-agent architecture novel. ChartLlama addresses multimodal chart understanding [5], while the cited Tsantekidis study concerns limit-order-book prediction, not candlestick-image recognition [6]. These tasks should not be treated as interchangeable validation of our chart reader.

Factor search and evaluation. The source catalogue of 101 Formulaic Alphas describes expressions with cross-sectional and other requirements [7]. Our 80 retained expressions are heuristic single-asset adaptations, with operators and deviations documented individually; five further formulas are custom to the implementation. Searching many candidates and choosing their orientation on the same history creates selection bias [8]. Nested selection and assessment [9], the Deflated Sharpe Ratio [10], and White's Reality Check [11] address related evaluation concerns. This study does not claim to have applied all those procedures: its auxiliary numeric study uses train-fit controls and a declared Holm family, while the native selector retains its audited defects.

Vietnamese sentiment and reproducibility. ViSoBERT was developed for Vietnamese social-media language understanding [12]. We use a distinct public 5CD-AI sentiment classifier derived from that backbone [13]; its identity does not establish accuracy on Vietnamese financial news. General financial sentiment work such as FinBERT [14] does not substitute for validating this classifier and corpus. Accordingly, we report the implemented scoring recipe, availability and provenance

rather than assuming financial-domain competence. Exact hosted-model replay poses an additional limitation: the text model's published description [15] does not identify the immutable weights served for an archived request. We retain observed aliases and fingerprints without treating them as weight attestations.

3. Forecasting task and signal construction

3.1. Decision, target and account action

Let d be the last observed session at a decision cutoff, $e = d + 1$ the planned entry session, and $L = 3$. The candidate LONG cycle buys at $Open_e$ and exits at $Close_{e+L-1} = Close_{d+3}$. Thus three observed bars include the entry bar, and exit is two observed sessions after entry. With brokerage $f = 0.0025$ and slippage $s = 0.001$ on each leg, the target is

$$r_d^{net} = \frac{Close_{d+3}}{Open_{d+1}} \frac{(1-f)(1-s)}{(1+f)(1+s)} - 1, \qquad y_d = \mathbf{1}[r_d^{net} > 0]. \tag{1}$$

UP means a strictly positive candidate LONG net return; zero belongs to DOWN. Gross return is a separate diagnostic on the same net-elicited predictions, not a replacement label or an additional inference experiment. The seven gross/net label disagreements are retained. Decision Close differs from entry Open at 65 of the 102 units, so a close-to-close label is not equivalent.

The agent chooses LONG or SHORT. In the long-only reference ledger, LONG allocates the account to the cycle and SHORT holds CASH; SHORT is not a short sale and a correct DOWN classification earns no bearish trading return. Account return therefore differs from the candidate-LONG target. Missing horizons are rejected rather than shortened. Canonical exports distinguish decision Close, entry Open and exit Close. A legacy field called `actual_prev_close` is not authenticated as decision Close merely because it sometimes equals entry Open; its unresolved semantic provenance is preserved in the legacy audit.

3.2. Native factor path and its causal defect

At each outer cutoff the native implementation receives a 600-session historical prefix and selects five of 85 candidates. Eighty are retained WorldQuant adaptations and five are custom formulas. The companion registry gives every source expression, handler, operator window, ties/fills and excluded expression; it does not assert mathematical equivalence to the source catalogue [7]. Among 21 exclusions, 17 involve industry neutralization, one market capitalization, and three price/volume omissions have no authenticated selection rationale. They are not all described as impossible cross-sectional translations.

Native candidate diagnostics use the trailing sample to estimate rank correlation (IC), sign orientation, directional accuracy and a signed-return Sharpe proxy, then rank a weighted composite. IC orientation and factor selection are consequently in-sample diagnostics, not independent factor discoveries. The normalization names include zscore–tanh, min–max and rank-based transforms; the actual whole-Series implementations, partial windows, denominator guards and NaN/zero fills matter more than their nominal names. Returns are implemented as percentage changes, and VWAP/ADV use OHLCV proxies rather than independently measured transaction VWAP or dollar volume.

The default score is $0.40\, m(|IC|) + 0.35\, m(Accuracy) + 0.25\, m(Sharpe)$, where m is min–max scaling across candidate diagnostics (a constant range maps to 0.5). Negative IC reverses the signal used for diagnostic accuracy and the signed-return proxy; displayed diagnostic rounding precedes ranking. Directional accuracy uses the strong-signal subset $|a| > 0.05$, with fallback conventions for insufficient support. The signed-return Sharpe proxy is not a realized long-only CASH ledger. The

nominal minimum-history check is not a causal warm-up trim: partially filled early rows can enter native diagnostics. Exact metric guards and tie/sort behaviour remain in the registry implementation rather than being replaced with an idealized formula.

The native row-wise causal gate fails. Counterexamples show that `delay().bfill()`, whole-Series normalization and the full-frame volume-availability branch can change earlier row values when later rows are extended or altered. In contrast, synthetic outer-cutoff probes found that poisoning data after the supplied 600-bar prefix did not change the native outputs. These are different properties: cutoff isolation in a fixture does not make all rows within the prefix causal, certify news PIT, or remove selection bias. We have not repaired the historical native inputs and then attributed the same saved LLM outcomes to a corrected mechanism.

Custom definitions are summarized in Appendix A. For example, native BFE uses $\tanh(1.5(\%B - 0.5)VolZ_5)$: negative volume z-score reverses the sign and zero cancels it. There is no volume-expansion or squeeze gate. Native registry and prototype/fallback paths have different preprocessing, including sample versus population SD and rounded technical variables. A selected-factor display rank is not a permanent factor ID. Six selected metadata slots on six main units disagree with handler descriptions; the affected actual prompts did not contain the literal wrong formula. Their semantic effect on the LLM has not been measured separately. Handler identity, descriptions and frozen outcomes remain distinguishable.

3.3. Sentiment recipe and availability

The archived local scoring recipe identifies `5CD-AI/Vietnamese-Sentiment-visobert` at revision `d7738b85200272f10ba8b048d80236c8c582b581`, with labels NEG/POS/NEU and article score $P(POS) - P(NEG)$. Tokenization uses an explicit 256-token limit, batch size eight, padding and truncation; tokenizer checks and saved probability/score checks do not constitute a new model evaluation. The released scorer is distinct from the ViSoBERT backbone citation [12, 13].

Availability requires at least three distinct identities with explicit accepted scorer provenance and finite scores in a 90-day cutoff window. Date-only publication is conservatively made available the following day; explicit availability metadata is retained. For eligible articles, the frozen aggregate is

$$S_d = \frac{\sum_i 2^{-age_i/30}\,[P_i(POS) - P_i(NEG)]}{\sum_i 2^{-age_i/30}}, \tag{2}$$

with age in days at the cutoff, hence a 30-day half-life. Missing sentiment is not an observed zero/neutral signal. The frozen provider supplies the aggregate/count report but empty article-detail and peer lists to the normalizer; its z-score and relative fields are therefore zero on this path, not independently observed peer information. The normalizer's reliability flag uses at least eight articles, distinct from the three-article availability rule. A false reliability flag alone does not mean the weighted report was missing. We do not substitute the prototype sentiment equations for these actual inputs.

Legacy inventory contains 180 article records and 177 unique identities, not 8,180 authenticated articles. All 180 legacy scorer identities are unknown; under the corrected strict eligibility/provenance rule none of the 180 legacy test cutoffs has three valid articles. Old runtime counts, duplicate memberships and article counts are retained separately from this strict audit. The reconstructed study and 154 locally rescored excerpts are a separate provenance scope. Actual prompt audits establish which sections were delivered, not that the news content was historically available or correctly interpreted. The authors have no independently labelled financial-news validation set. Social-media domain mismatch and unauthenticated content/publication availability preclude financial-domain accuracy and PIT-certified sentiment claims.

4. Implemented multi-agent system

4.1. Roles and actual information flow

Figure 1 describes the implementation evaluated here. Indicator Agent computes and renders five deterministic TA-Lib-based votes: MACD, RSI, Stochastic, Williams %R and ROC. They are momentum/trend indicators, not five volatility oscillators. Its backtest node returns a report and messages, without calling an LLM or forwarding the declared state schema's complete numeric arrays. Bollinger variables can be computed on the alpha path; BB/ADX state declarations do not create two additional Indicator votes. Appendix A specifies the ordered boundaries and missing-data convention [16, 17].

The Alpha role supplies native selected-factor and sentiment context. In the evaluated backtest its numerical/report path bypasses an additional alpha-reasoning LLM invocation. Pattern and Trend read their respective cached chart images through the vision model, and Decision receives condensed Indicator, Pattern and Trend reports plus the enabled factor/sentiment sections. Indicator, Pattern, Trend and Decision roles, shared state, chart tools and trendline procedure reuse or modify QuantAgent code; the upstream MIT notice is preserved in the repository [1]. Domain adaptation, input processing and the evaluation protocol should be distinguished from the reused architecture.

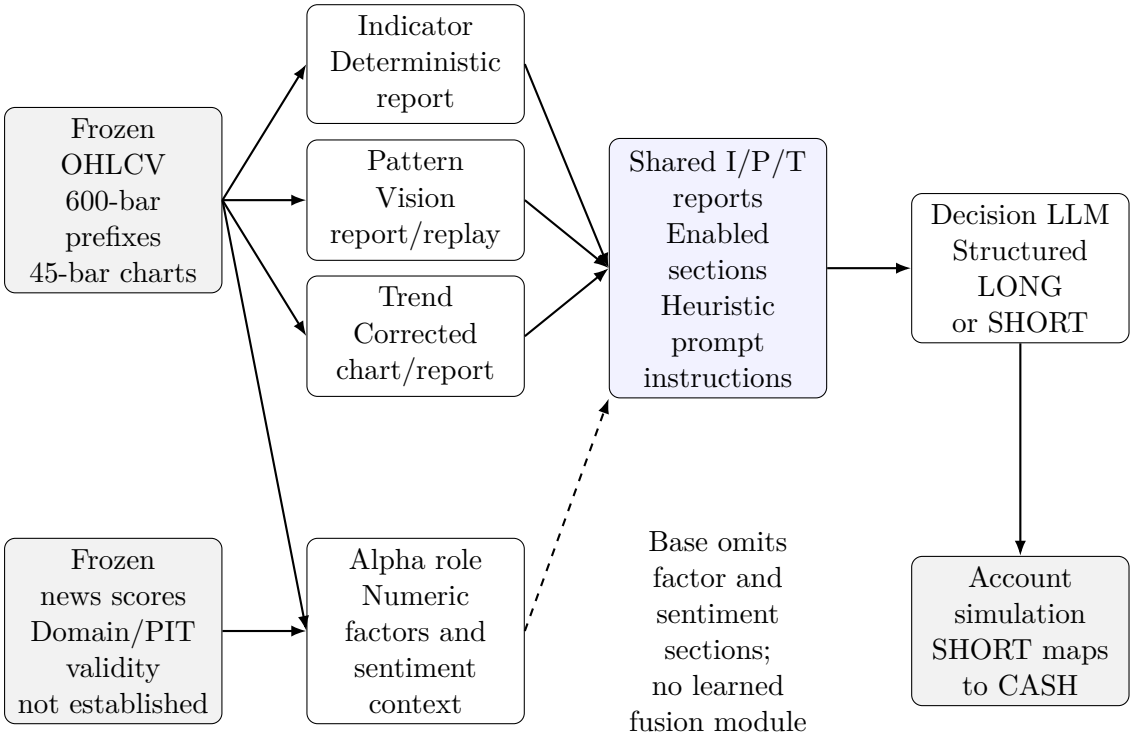


Figure 1: Implemented information flow and shared-report comparison. Aggregation uses prompt instructions rather than a learned fusion module. Indicator and alpha backtest reports do not require additional reasoning-model calls.

4.2. Charts and aggregation

Figure 2 shows the earliest chronologically eligible main unit, chosen by date then symbol/unit ID without inspecting outcomes. Images are exact archived PNGs: Pattern is replayed from its original request, while Trend uses the corrected legend. Candle bodies are green when Close exceeds Open and red otherwise, including a doji; this is an Open–Close body rule, not the exchange price board's



Figure 2: Actual Pattern and corrected Trend inputs for CMG on 3 July 2023, the first chronologically eligible main unit. Bodies compare Close with Open; the Trend legend maps blue to support and red to resistance. These are historical inputs, not synthetic fixtures or evidence of validated visual understanding.

reference-price rule. Support is blue and resistance red. The original claim of green bearish/red bullish candles following Vietnamese market convention is withdrawn.

The renderer audit matched 204 images across 102 units byte for byte and pixel for pixel, and 240 actual vision requests matched their image caches and Vietnamese templates. It verified body geometry, colour and legend handles. It did not establish what the vision model attended to. In particular, the Vietnamese word *xanh* is less colour-specific than the corrected English "blue" wording. Inherited high-frequency/K-line prompt language also remains distinguishable from the daily-bar application. Deterministic consistency flags can catch obvious output contradictions but cannot certify visual grounding.

Reports are distilled and capped before Decision. The observed successful Decision prompts are at most 4,187 characters, below the 6,500-character guard. Priority and conflict instructions prefer price/chart evidence, allow quantitative factors to challenge it, and give sentiment lower priority. These are prompt heuristics, not learned/calibrated weights, an explicit knowledge representation, or guaranteed conflict resolution. The system performs prompt-level report aggregation; no separate experiment validates a new reasoning or knowledge-fusion algorithm. Complete actual prompts and consistency rules belong to the companion archive, not lengthy article appendices.

Base retains the same upstream report content and general instructions but removes both alpha and sentiment sections. Thus general instructions may still mention those concepts. Main/upstream prompts use a compact factor representation, whereas the factorial panel delivers five explicit value rows. We retain this representation difference in component comparisons rather than assuming identical information delivery across prompt families.

5. Historical evaluation protocol

5.1. Snapshot, calendar and scope amendment

The universe is purposive: BHN, CMG, FPT, HVN, MBB, MWG, VCB, VJC and VNM. The reconstructed vnstock/VCI price snapshot covers observed vendor sessions from 2020 through September 2026. It is not an authenticated submission-time snapshot. Contradictory BHN OHLC data trigger the frozen valid-suffix policy; insufficient subsequent warm-up excludes BHN from three early blocks. No price is corrected to improve a result. This leaves 102 eligible cells of the 12-date by nine-symbol grid in Figure 3.

Six blocks cover 2023 H2, 2024 H2, 2025 H1, 2025 Q4, 2026 Q2 and 2026 Q3, with two scheduled dates per block. These are twelve selected dates, not a continuous multi-year daily backtest. Cross-symbol observations share calendar shocks. The 51 eligible symbol–block scopes contain chronological train/validation/test partitions: every train label exits before validation starts, and every validation label exits before the first test decision. Cutoffs require the frozen history length; label spans touching the next partition are purged.

The reduced scope was amended after some inference to complete a bounded historical study. Its 375 registered Decisions include 193 explicitly reused successful executions and 182 further successes. The factorial subset was selected by metadata (symbol/date round-robin), not prediction quality, but this does not make the overall amendment preregistered. The larger legacy grids remain incomplete/superseded. Table 1 distinguishes shared r0 calls, additional realizations and the separate factorial panel; their observations are not summed as independent markets.

5.2. Comparators, repeats and placebo

All nine classifiers use the same 102 main units and net target. CASH and always-LONG are constant predictions; always-LONG closes and pays costs every cycle. Train-majority uses purged

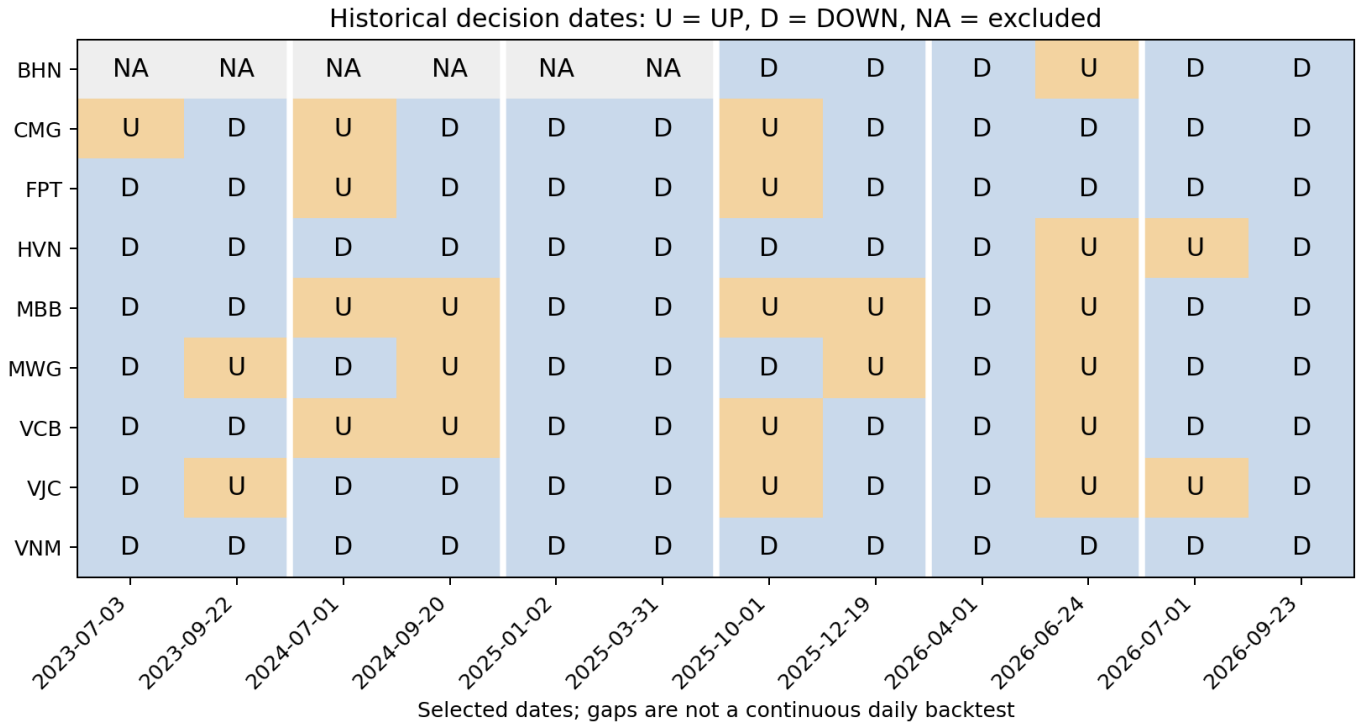


Figure 3: Selected historical decision dates and eligibility across nine equities. Missing BHN cells remain excluded rather than being filled with CASH predictions. Labels are shared evaluation truth, not a visualization selected from Full's successful forecasts.

Table 1: Evaluation scopes and prompt variants in the frozen historical study. Shared main/upstream executions are counted once.

A. Recorded supports

Scope	N/exec.	Symbols	Dates	Blocks	UP/DOWN	Exec.	Decisions
Main	102	9	12	6	25/77	1	204
Upstream	9	9	3	3	2/7	3	54
Factorial	9	8	4	3	2/7	3	135

B. Prompt sections

Variant	Factors	Sentiment	Representation
B	–	–	Shared price reports
A	Current	–	Five-row factorial
S	–	Current	Factorial
AS	Current	Current	Compact main / five-row factorial
P	Past donor	Current	Five-row factorial

Main uses B/AS; upstream uses B/AS; factorial uses B/A/S/AS/P. Class counts are per execution. Upstream includes 18 main r0 calls: 204+54+135-18=375 unique Decisions. P preserves current names/schema/missingness and sentiment, replacing factor values/signals/skill metadata by one past train-only rank-vector donor.

training labels, choosing CASH on a tie. Momentum predicts LONG for a strictly positive trailing 20-session return. Logistic-price uses numerical price features; logistic-matched adds the recorded factor/sentiment/missingness inputs. Imputation, scaling, regularization and decision threshold selection follow chronological train/validation rules; final logistic refits use matured train plus validation only. We reuse their saved predictions rather than refitting after observing outcomes. The fixed factor-vote comparator uses training-fitted medians/orientation and differs from the native top-five policy and the auxiliary numeric-control study.

Matched numerical inputs do not equal chart images, report wording or pretrained knowledge. Full–Base differences likewise include prompt structure, report length and propensity to choose CASH. The upstream panel has nine units, B/AS and three regenerated upstream realizations with a Decision execution per realization. The factorial panel has nine different-support units, B/A/S/AS/P and three Decision executions with fixed upstream. Both have two UP/seven DOWN per execution, but only the former covers nine symbols. We do not infer main-grid variability or separate upstream/Decision variance components from their SDs.

P retains current factor names, headings, schema/missingness and sentiment, replacing scalar values, signals and skill fields using a predetermined earlier train-only rank-vector donor. Donor label exits precede validation/test. Rank mapping is not name-matched covariance preservation, a placebo distribution or exposure matching. Five explicit rows reach each A/AS/P factorial prompt. Main/upstream compact prompts are a separate representation family. Interaction is defined as $AS - A - S + B$; the observed best-single gap $AS - \max(A, S)$ is a different, post-hoc diagnostic and not an eighth factorial contrast.

5.3. Metrics, dependence and execution assumptions

BA averages UP and DOWN recall; MCC and both-class precision/recall accompany ordinary accuracy and LONG rate. Pooled metrics retain actual class supports. Symbol macro means are unweighted and report planned/defined denominators. Missing class support or a degenerate MCC denominator yields NA, not zero. Main contrasts use 10,000 paired whole-block resamples, seed 20261102, retaining dates, cross-section, variants and eligibility together. Repeat diagnostics jointly resample recorded blocks and paired execution indices. Intervals condition on finite draws; undefined counts remain visible. Circular block lengths, leave-one-block-out and Bartlett HAC at lags 0/1/2 are sensitivity diagnostics. HAC uses calendar-aligned accuracy differences in fixed-universe panels, with a $t(n_{dates} - 1)$ reference; few clusters, irregular gaps and post-inference additions prevent calibrated significance or power claims.

Each of 51 symbol–block accounts starts with VND50 million, without cross-symbol capital competition. LONG follows Eq. 1; CASH stays flat. Continuous hold buys once from the first entry through the last scheduled exit, including gaps, and is not a tenth classifier. Wealth includes pending receivables and residual shares at recorded mark-to-market prices. Daily-close drawdown uses an initial-capital anchor. Where supplied in companions, daily Sharpe uses population SD and a 252-session convention with zero risk-free rate; the retained Sortino convention uses SD of negative daily observations, not downside RMS. Annualised cycle ratios are NA for the irregular/gapped main schedules, and are not presented as investment-quality estimates from a handful of trades.

The reference assumes fractional fills at recorded prices and two-leg fees/slippage. Shares are assumed sellable in the afternoon of T+2 conditional on allocation and fill; sale proceeds are pending until the specified settlement phase. This convention follows the archived settlement source [18]. L=3 is a fixed observed-bar protocol choice, not an optimized horizon. Daily OHLC cannot authenticate opening-auction queues, price-limit fills, partial fills or correct volume units. All twelve sensitivities, including pending cash, advances, cost changes, sell charge, no-fill/delay proxies and lot/capacity

assumptions, remain in Appendix B. Venue/calendar review flags, including ten HVN unit flags, are retained rather than converted to certified fills or synthetic CASH.

5.4. Models and reproducibility

Archived SDK aliases are `openai/gpt-oss-20b` for text and `qwen/qwen3.8-27b` for vision. Configured temperature zero appears as 10^{-8} in observed SDK metadata; no hosted decoding seed is recorded. Text/vision output caps are 2,048/1,024 tokens. Retry/checkpoint records preserve individual attempts and completed outputs. Aliases and 40 response fingerprints do not authenticate an immutable weight revision or historical provider catalogue; 91 original Pattern requests lack epoch/provider-org fields and remain unknown. Offline replay reproduces saved-prediction metrics, not a fresh hosted inference experiment.

The closure contains 649 attempts, 615 successes and 34 errors, including 120 original Pattern calls replayed as shared dependencies. Observed usage totals 1,063,328 prompt and 340,736 completion tokens. Error usage/latency and monetary billing are unknown. Pattern tokens/latency describe original calls, not replay lookup cost; successful callback elapsed times are not wall-clock runtime. Full accounting, pins and hashes are in repository artifacts rather than an implementation appendix.

6. Results

6.1. Main class-aware comparison

Table 2 uses all 102 main units, not the 180-point legacy study. Full accuracy is 65.69%, Base 60.78%, and CASH/train-majority 75.49%. Because 77/102 outcomes are DOWN, ordinary accuracy alone favours conservatism. Full BA is 52.96%, Base 49.71%, and logistic-price 55.61%; the Full–Base BA difference is 3.25 percentage points. All comparators, including less favourable ones, remain in the table. Undefined constant-rule MCC is NA.

Full and Base each identify seven of 25 UP outcomes. Full issues 24 LONG forecasts, with 17 false positives, versus Base's 29 LONG forecasts and 22 false positives. Thus the observed BA difference accompanies improved DOWN recall rather than additional captured UP cases. This pattern is compatible with a shift towards CASH; it does not identify why the shift occurred or establish information-specific alpha. Neither a higher score than Base nor a smaller loss than an always-invested rule is sufficient evidence of profitability or broad superiority.

Pooled and macro values are different estimands. BHN contributes six eligible units, the other eight symbols twelve; missing class support reduces some macro denominators. Table B.1 reports Full/Base by symbol. Companion S1 includes all 81 symbol–classifier rows, both-class metrics and all pooled/macro denominators, so comparator reporting is not restricted to selected symbols or the main table's aggregate.

6.2. Account outcomes and exposure

Full mean net account return is -0.797%, versus Base -0.750% and CASH zero (Table 3). Full therefore does not even outperform Base on this main economic mean, despite its higher BA. Always-LONG cycles and continuous hold have different costs and gap exposure, and neither is substituted for the other. The held-session fractions include the long CASH gaps between the two cutoffs in each block; their low values should not be confused with LONG percentages over scheduled decision units.

Returns and drawdowns summarize separate accounts, not one portfolio, total-return index or certified executable strategy. Risk ratios cannot recover evidence of profitability from sparse trades. The main schedules have 57 observed sessions between cutoffs, so pending-cash/on-demand-advance

Table 2: Class-aware performance on the same 102 historical market units. Macro summaries weight symbols equally and retain undefined denominators.

A. Pooled (percent except MCC)

Strategy	Acc.	BA	MCC	P_U	R_U	P_D	R_D	LONG
Full	65.69	52.96	0.060	29.17	28.00	76.92	77.92	23.53
Base	60.78	49.71	-0.005	24.14	28.00	75.34	71.43	28.43
CASH	75.49	50.00	NA	NA	0.00	75.49	100.00	0.00
Train-majority	75.49	50.00	NA	NA	0.00	75.49	100.00	0.00
Always-LONG	24.51	50.00	NA	24.51	100.00	NA	0.00	100.00
Momentum-20	49.02	47.32	-0.046	22.45	44.00	73.58	50.65	48.04
Logistic-price	67.65	55.61	0.114	33.33	32.00	78.21	79.22	23.53
Logistic-matched	60.78	52.42	0.044	27.27	36.00	76.81	68.83	32.35
Factor vote	51.96	51.97	0.034	26.00	52.00	76.92	51.95	49.02

B. Unweighted symbol macro summaries

Strategy	BA (%)	Defined	MCC	Defined
Full	55.38	8/9	0.08	8/9
Base	51.48	8/9	0.00	8/9
CASH	50.00	8/9	NA	0/9
Train-majority	50.00	8/9	NA	0/9
Always-LONG	50.00	8/9	NA	0/9
Momentum-20	50.67	8/9	-0.01	8/9
Logistic-price	59.00	8/9	0.17	7/9
Logistic-matched	56.55	8/9	0.08	8/9
Factor vote	46.86	8/9	-0.04	8/9

$N = 102$, UP/DOWN=25/77. P/R denote class precision/recall. NA is undefined, including MCC for constant predictions. Train-majority is fit on purged train labels. Logistic-matched uses common numerical market inputs, not identical chart/text representations. Full per-symbol/class metrics are supplied in companion S1; hold has no classification prediction.

equivalence on these accounts provides little evidence about rapid capital reuse. All twelve economic sensitivities are reported in Table B.2. The all-entries-missed scenario gives zero but is not a mathematical worst-return bound; other assumptions may introduce unfilled entries, delays, pending balances or residual shares. Predictions and classification labels are unchanged across these sensitivities.

6.3. Repeated executions and factorial contrasts

Table 4 retains all three executions, sample SD, range and cutoff-level prediction disagreement summaries. The two panels have nine market units each but different symbols, dates and prompt families. Their means and SDs are not transferable to the broad 102-unit main study. Zero metric SD does not imply identical predictions: factorial S has unchanged aggregate BA but nonzero disagreement. Return means use nine upstream or eight factorial symbol–block accounts per variant/execution.

On the factorial panel the AS–B BA mean is −9.52 pp and AS–P is −11.90 pp. Interaction $AS - A - S + B$ is +14.29 pp, whereas the gap of mean BA between AS and the observed best

Table 3: Account outcomes under the fractional reference ledger. Means summarize 51 separate symbol–block accounts per strategy and do not establish executable portfolio returns.

Reference ledger

Strategy	Return (%)	MDD (%)	Held (%)	Orders
Full	-0.797	1.37	2.35	0.94
Base	-0.750	1.45	2.84	1.14
CASH	0.000	0.00	0.00	0.00
Train-majority	0.000	0.00	0.00	0.00
Always-LONG	-3.000	4.52	10.00	4.00
Momentum-20	-1.636	2.58	4.80	1.92
Logistic-price	-0.683	1.18	2.35	0.94
Logistic-matched	-1.105	1.60	3.24	1.29
Factor vote	-1.457	2.49	4.90	1.96
Continuous hold	-2.127	12.94	100.00	2.00

Each row averages 51 independent-capital symbol–block accounts, not a portfolio. Held is the fraction of observed calendar sessions held, including scheduled CASH gaps; it differs from the LONG rate. Continuous hold buys once and sells once across gaps. Dividends/corporate actions and fills are not certified. Annualised cycle Sharpe/Sortino are omitted for irregular/gapped schedules.

single component is zero. These are different quantities. All seven descriptive BA family envelopes contain zero; only 6,986 of 10,000 BA draws retain both classes and 3,014 are undefined. Conditional intervals are not zero-imputed or advertised as calibrated familywise coverage. All factorial mean account returns are negative.

The upstream AS–B mean BA difference is +4.76 pp, with conditional percentile interval $[0, 10]$ pp. This higher-performing small panel does not replace the main or factorial outcomes. Actual shared reports and paired prompt contents match their frozen parents, but 40 of 147 groups have a successful-call order different from the listed job order, and nine span execution epochs. Routing/quota/resume did not implement full counterbalancing. We retain every outcome and disclose the resulting time/order confounding rather than selecting a clean-looking subset or repairing it retrospectively.

6.4. Temporal and dependence diagnostics

Table 5 keeps the all-102 primary cohort and two overlapping structural cohorts. The primary Full–Base BA interval is $[-1.18, 8.13]$ pp. The common-eight cohort has 96 units over twelve dates; recent-nine has 54 over six dates. They are sensitivity views, not independent replications or grounds for choosing a more favourable headline. The six-block sample includes an all-DOWN early-2025 block: within-block BA without both classes is undefined rather than fabricated.

Past Close/SMA60 contexts distinguish bull, bear and sideways observations. These are per-security descriptions, not validated market-wide index regimes or a regime-conditioned trading policy. Additional volatility and turnover-proxy thresholds are fitted on purged training data, but their post-inference stratification remains descriptive. Close times Volume is in vendor units, not measured spread/depth liquidity. Exposure based on three target sessions differs from whole-calendar exposure with gaps.

Calendar-aligned HAC accuracy effects for common-eight/recent-nine are +3.125/ + 9.259 pp. At lags 0/1/2, common-eight SEs are 0.026042/0.023146/0.020172 with p-values 0.255346, 0.204107 and 0.149610; recent-nine SEs are 0.040713/0.047112/0.045710 with p-values 0.072051, 0.106554 and

Table 4: Execution variability and paired contrasts on two distinct nine-unit panels. Repeated executions do not add independent market dates.

A. Three executions: mean (sample SD)

Panel/variant	Acc. (%)	BA (%)	BA range	Disagr. (%)	Return (%)
U B	85.19 (16.97)	90.48 (10.91)	[78.57, 100.00]	22.22	0.37 (0.38)
U AS	92.59 (6.42)	95.24 (4.12)	[92.86, 100.00]	7.41	0.50 (0.22)
F B	44.44 (11.11)	28.57 (7.14)	[21.43, 35.71]	14.81	-1.70 (0.55)
F A	29.63 (6.42)	19.05 (4.12)	[14.29, 21.43]	22.22	-2.03 (0.31)
F S	22.22 (0.00)	14.29 (0.00)	[14.29, 14.29]	22.22	-2.34 (0.22)
F AS	29.63 (16.97)	19.05 (10.91)	[7.14, 28.57]	22.22	-2.23 (0.63)
F P	48.15 (6.42)	30.95 (4.12)	[28.57, 35.71]	14.81	-1.42 (0.33)

B. BA contrasts in percentage points

Contrast	Mean	SD	Range	Percentile	Family env.
U AS-B	4.76	8.25	[0.00, 14.29]	[0.00, 10.00]	NA
F AS-B	-9.52	10.91	[-21.43, 0.00]	[-22.22, 9.52]	[-37.14, 18.10]
F A-B	-9.52	4.12	[-14.29, -7.14]	[-20.37, 14.29]	[-37.14, 18.10]
F AS-S	4.76	10.91	[-7.14, 14.29]	[-6.67, 24.07]	[-22.86, 32.38]
F S-B	-14.29	7.14	[-21.43, -7.14]	[-29.63, 10.00]	[-41.90, 13.33]
F AS-A	0.00	12.37	[-14.29, 7.14]	[-20.00, 11.11]	[-27.62, 27.62]
F AS-P	-11.90	14.87	[-28.57, 0.00]	[-28.57, 4.76]	[-39.52, 15.71]
F interaction	14.29	14.29	[0.00, 28.57]	[-16.67, 38.89]	[-13.33, 41.90]

U: nine symbols/three dates/three blocks; F: eight symbols/four dates/three blocks. Each panel has nine units and UP/DOWN=2/7 per execution. Returns average nine U or eight F accounts per variant/execution. BA intervals retain 6,986 finite and 3,014 undefined draws of 10,000; envelopes are descriptive, not calibrated simultaneous coverage. U AS-B is separate from the seven F contrasts. Forty/147 groups deviate from listed variant order; nine span execution epochs.

0.098654. These are accuracy, not BA, effects on twelve/six selected dates. Companion S4 retains all six rows, the calendar masks, all regime strata, block-length sensitivities and leave-one-block-out diagnostics. Few clusters and irregular dates prevent reliable significance, null-effect or calibrated-coverage conclusions.

6.5. Auxiliary numeric selector controls

Table 6 is a separate numeric study, not an LLM factor-selector ablation. Causal adapters and train-only eligibility, imputation, scaling, orientation and ranking are fitted once per symbol–block scope, then retained through validation/test. Fixed custom five, dynamic train top-five, the same IDs without sign flip, and all twenty random eligible-five trials produce 2,346 predictions on the same 102 units. No random seed is selected after observing results.

Dynamic-minus-fixed, dynamic-minus-random-family and dynamic-minus-no-flip BA differences are respectively +15.84, +7.33 and +6.44 pp. Exact six-block two-sided sign-flip values are 0.0625, 0.125 and 0.53125; Holm correction over all 23 declared hypotheses gives adjusted $p = 1$ throughout. Unadjusted percentile intervals and the entire random dispersion remain visible. Positive point differences do not establish superiority, and the no-flip contrast does not identify sign-flip value inside the LLM pipeline.

Table 5: Overlapping temporal cohorts and observed trend strata. Small-block intervals and calendar-aligned HAC diagnostics are descriptive rather than calibrated significance tests.

A. Cohorts: BA percent; difference/interval in pp

Cohort	N	Dates	Blocks	Full	Base	Δ	Interval
All	102	12	6	52.96	49.71	3.25	[-1.18, 8.13]
Common eight	96	12	6	50.69	48.61	2.08	[-1.20, 5.26]
Recent nine	54	6	3	53.08	46.67	6.41	[0.00, 12.50]

B. Observed per-security trend contexts (percent)

Trend	N	UP/DOWN	Full BA	Base BA	Full LONG	Base LONG
Bear	35	5/30	43.33	41.67	11.43	14.29
Bull	33	9/24	47.92	53.47	36.36	39.39
Sideways	34	11/23	59.49	46.25	23.53	32.35

Cohorts overlap and are not pooled as extra samples. Trend uses past Close/SMA60 with strict +/-2% thresholds; it is not an index-regime policy. Primary intervals retain both dates, full cross-sections and pairing within blocks. HAC concerns accuracy rather than BA and is reported for both fixed-universe panels at lags 0/1/2 in S4.

Native diagnostics separately account for 102 cutoffs, 8,670 candidate opportunities and 510 selected slots. WQ42 occurs at 52/102 cutoffs (52/510 slots); adjacent nonzero IC signs flip 1,282 times over 7,894 eligible transition pairs. These are dependent in-sample diagnostics, not out-of-sample alpha stability. Complete per-factor frequency, eligible denominators, IC signs, windows and all 21 exclusions are retained in the registry companion. Numeric controls do not repair native row-wise noncausality, provide the missing fixed/random LLM experiment, or certify original-expression equivalence.

Table 6: Train-fit numeric selector controls using separate causal adapters. These auxiliary results neither repeat the native LLM selector experiment nor repair its row-wise causal defect.

A. Pooled numeric results; random mean (SD)

Strategy	Trials	Acc. (%)	BA (%)	MCC	Random BA range
Fixed custom five	1	49.02	39.22	-0.193	–
Dynamic train top-five	1	62.75	55.06	0.093	–
Same IDs, no flip	1	50.98	48.62	-0.024	–
Random eligible five	20	51.27 (5.32)	47.74 (5.38)	-0.039 (0.094)	[37.97, 57.17]

B. Three primary contrasts; multiplicity family = 23

Contrast	ΔBA (pp)	Percentile interval	Exact p	Holm p
Dynamic - fixed	15.84	[6.82, 25.22]	0.0625	1.0000
Dynamic - random mean	7.33	[0.57, 13.75]	0.1250	1.0000
Dynamic - no flip	6.44	[-10.93, 20.26]	0.5312	1.0000

$N = 102$, UP/DOWN=25/77, 51 train-fit scopes. Numeric adapters/preprocessing are separate from the native LLM pipeline. All 20 random trials and all 23 contrasts remain in S3. Exact tests have 64 six-block sign assignments; percentile intervals are unadjusted, Holm uses the full 23-hypothesis family. No contrast rejects at 0.05.

7. Discussion and limitations

What the comparisons establish. The archived implementation yields identifiable forecasts and replayable metrics, but its Full–Base contrast changes a bundle of inputs and prompt structure. Main/upstream compact reports differ from explicit factorial values. Matching report parents or numerical inputs does not match text/image representations, pretrained knowledge, exposure or semantic use. The rank-vector placebo tests one predetermined past replacement, not a distribution of independent placebo treatments. Consequently, these results describe recorded behaviour and do not isolate causal alpha, sentiment or selector value. Byte-level duplication in legacy artifacts is audited rather than treated as proof that historical branch labels were correct; those legacy ablations no longer support component attribution.

Causal construction and domain validity. The native row-wise causal failure in Section 3 remains unresolved in the frozen study. Its in-sample ranking/orientation and heuristic expression translations limit any factor-discovery interpretation. The numerical controls use a separate causal path, so their passing probes cannot certify the earlier native inputs. Similarly, saved scorer probabilities, a pinned local checkpoint and strict date rules cannot authenticate the original news content or publication availability. No independent financial-domain labels, agreement/adjudication or domain test performance are available. Missing sentiment is not evidence that sentiment was neutral, and rescoring is not validation. Repairing these defects for a new experiment would require versioned inputs and a new protocol, not rewriting old predictions.

Sampling, dependence and hosted variability. The universe is purposive and potentially survivor-biased; point-in-time listing/delisting/suspension membership is unverified. All outcome dates precede the November 2026 freeze, so model training contamination remains possible even if a named model was publicly released before some test dates. Twelve dates and six blocks, plus three executions on each small panel, do not establish generalization, effective sample size or calibrated uncertainty. Repeats, correlated symbols, overlapping cohorts and repeated bootstrap draws are not extra independent markets. Order deviations, epoch/routing changes and unknown immutable served weights further confound distributional comparisons. The historical scope amendment makes this an exploratory study rather than an untouched confirmatory evaluation.

Economic and visual scope. Negative account means do not demonstrate a deployable strategy. Reference fills, fee/slippage, settlement phases, lot/capacity limits and volume units are assumptions or proxies, while corporate-action adjustments, price-limit queues, actual spread/depth and partial fills are unauthenticated. Separate account averaging omits portfolio capital competition. L=3 is not optimized. Visual byte/pixel/legend integrity and deterministic consistency flags do not show that the VLM read the claimed candle or trendline; matched counterfactual charts and local-evidence annotation would be needed to test that stronger claim. Prompt precedence is not a learned or guaranteed conflict-resolution mechanism.

Next evidence and responsible interpretation. A prospective versioned study should repair native row causality, repeat selection/preprocessing inside chronological training, and preserve a later untouched evaluation interval. Candidate search can be assessed with procedures such as White's Reality Check or a Deflated Sharpe analysis [11, 10]; neither is claimed as completed here. Independent financial-news labels and human visual-evidence annotations are needed before domain/semantic reliability claims. More calendar coverage and controlled repeated hosted calls would be required for stable model/component conclusions; executable portfolio validation needs separately authenticated market and settlement information. Public release is also conditional on rights and a real archive

identifier. These are outstanding requirements, not benefits inferred from the current experiment. The implementation is research decision support and should not be represented as profitable trading advice.

8. Conclusion

We evaluated a specific multi-agent LLM implementation for Vietnamese equities on a frozen historical scope, with common-support controls, archived calls and explicit label/account contracts. On 102 units, Full BA is 52.96% versus Base 49.71%; ordinary accuracy is strongly affected by class imbalance, a price-only logistic comparator has BA 55.61%, and Full's mean account return is -0.797%. Repeated panels reveal variability and do not establish stable component advantages. The contribution is a reproducible record of this implementation's observed behaviour and its limits, rather than evidence of causal alpha or deployable profitability. Native row-wise noncausality, in-sample selector diagnostics, missing financial-news/PIT validation and order/epoch confounding remain disclosed. A corrected prospective study and stronger independent validation are needed before restoring the claims those limitations currently prevent.

Declaration of competing interest

Authors state no conflict of interest.

Data and code availability

The project repository records the implementation and earlier result artifacts. The final reproducibility package and scientific companions S1–S4 have been prepared as a private local draft. Public distribution rights and a persistent archive identifier remain pending; no DOI or public availability of the full package is asserted. OHLCV/news snapshots are reconstructed research inputs rather than authenticated submission-time data. Distribution remains subject to source terms, news copyright, model licences and permissions; the upstream QuantAgent MIT notice does not grant rights over all additions or data.

Companion S1 contains `S1_classification.csv`, `S1_per_class.csv` and `S1_macro.csv`; S2 contains all executions, contrasts and disagreements; S3 contains numeric metrics, random trials and the full family; S4 contains calendar, HAC, block sensitivity, regimes and dependence diagnostics. Exact source paths and hashes are recorded in the manuscript release. The private raw package separately retains inputs, messages/images, responses/errors, parse/checkpoints, pins and provenance.

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Appendix A. Signal definitions and implementation distinctions

The tables retain scientific definitions needed to interpret the implemented reports. Full Python, operator windows/guards, expression registry and complete prompts remain in the companion repository/package. Declared state slots do not imply their numerical arrays reached Decision; the actual Indicator path delivers report text and messages.

Table A.1: Indicator signal rules and missing-data conventions.

Five deterministic votes

Signal	Priority and boundaries	Qualification
MACD	Previous strict/current inclusive crossing first; otherwise histogram sign and change.	12/26/9; no Stochastic histogram.
RSI	≥ 70: bear; ≤ 30: bull; $55 \leq x < 70$: bull; $30 < x \leq 45$: bear.	45–55 uses change against the third retained sample.
Stochastic	Crossing precedes ≥ 80 bear / ≤ 20 bull; interior fallback retained.	Legacy equality/zero-D truthiness is disclosed.
Williams %R	≥ -20: bear; ≤ -80: bull; otherwise retained-sample direction.	Not a volatility vote.
ROC	$> +0.5\%$: bull; $< -0.5\%$: bear; equality neutral.	Thresholds follow four-decimal rounding.

None/nonfinite values are removed, zero retained; rounding is to four decimals before classification. Retained samples need not be consecutive if missing. Missing/error gives a neutral vote within denominator five. Directional plurality, ties MIXED, and confidence are heuristics, not calibrated probabilities.

Table A.2: Custom native factor definitions and implementation distinctions.

Before global normalization and IC orientation

ID	Native handler expression	Distinction
FDM	$\tanh(0.5\, a_t V_t / ADV_{20,t})$	Native uses adjusted one-bar return, not the MACD display template.
SFA	$\tanh(1.5 \tanh(H_t/(0.005C_t + \epsilon)) + 0.5a_t)$	Native uses MACD histogram and return, not sentiment; prototype has a distinct sentiment branch.
LVR	$\tanh(-0.8\, z_{20}((C_t - SMA_{5,t})/C_t))$	Rolling mean/SD conventions and partial windows are implementation-specific.
BFE	$\tanh(1.5(\%B_t - 0.5)\, VolZ_{5,t})$	Negative VolZ reverses the sign; zero cancels. No squeeze or volume-expansion gate.
OFE	$\tanh(3a_t\, CP_t)$	OHLCV proxy, not measured signed order flow.

a_t is one-bar pct-change divided by rolling 20-bar sample SD (minimum five); H_t is MACD(12,26,9) histogram; CP_t is close location relative to the bar midpoint/range; ADV is mean Volume, not dollar volume. Guarded denominators and fills follow the registry. BFE registry sample-SD preprocessing differs from the rounded/population-SD prototype. Six selected metadata slots differ from their handler description, without the literal wrong formula in the affected prompts. A display rank of four does not identify BFE. Full expressions, guards and counterexamples are in the companion registry.

Appendix B. Additional account and symbol results

These tables use the same final supports and unchanged predictions as the main results. Supplement S1 gives every baseline by symbol and both classes; S2 contains all execution rows/contrasts; S3 the complete numeric/random family; S4 all temporal/regime/HAC diagnostics. They are scientific companion tables, distinct from raw execution logs and implementation files. No additional market observation is created by these alternative tabulations.

Table B.1: Full and Base symbol-level results on their common historical support.

Percent except MCC

Symbol/variant	UP/DOWN	Acc.	BA	MCC	R_U	R_D
BHN/Full	1/5	100.00	100.00	1.000	100.00	100.00
BHN/Base	1/5	66.67	80.00	0.447	100.00	60.00
CMG/Full	3/9	66.67	44.44	-0.174	0.00	88.89
CMG/Base	3/9	75.00	61.11	0.258	33.33	88.89
FPT/Full	2/10	66.67	40.00	-0.200	0.00	80.00
FPT/Base	2/10	50.00	30.00	-0.316	0.00	60.00
HVN/Full	2/10	66.67	60.00	0.158	50.00	70.00
HVN/Base	2/10	66.67	60.00	0.158	50.00	70.00
MBB/Full	5/7	50.00	48.57	-0.029	40.00	57.14
MBB/Base	5/7	58.33	55.71	0.120	40.00	71.43
MWG/Full	4/8	58.33	56.25	0.120	50.00	62.50
MWG/Base	4/8	50.00	50.00	0.000	50.00	50.00
VCB/Full	4/8	58.33	50.00	0.000	25.00	75.00
VCB/Base	4/8	50.00	37.50	-0.316	0.00	75.00
VJC/Full	4/8	58.33	43.75	-0.213	0.00	87.50
VJC/Base	4/8	50.00	37.50	-0.316	0.00	75.00
VNM/Full	0/12	83.33	NA	NA	NA	83.33
VNM/Base	0/12	83.33	NA	NA	NA	83.33

BHN contributes six eligible units, other symbols twelve; NA retains missing-class denominators. S1 additionally contains all nine classifiers for every symbol and both-class precision/recall/F1, with pooled and macro summaries.

Table B.2: All twelve account-execution sensitivities on unchanged predictions.

Means over 51 accounts per variant

Scenario	Base (%)	Full (%)	ΔFull (pp)	Stock	Pending
reference	-0.750	-0.797	0.000	0	0
pending_t2	-0.750	-0.797	0.000	0	0
advance_on_demand	-0.750	-0.797	0.000	0	0
advance_auto	-0.803	-0.840	-0.043	0	0
lower_cost	-0.526	-0.612	0.185	0	0
higher_cost	-1.249	-1.211	-0.414	0	0
sell_charge_10bp	-0.805	-0.843	-0.046	0	0
no_fill_proxy	-0.659	-0.803	-0.006	0	0
delayed_proxy	-0.621	-0.766	0.030	0	0
lot100_cap1pct	-0.602	-0.745	0.052	0	0
combined	-1.074	-1.166	-0.369	0	0
all_entries_missed	0.000	0.000	0.797	0	0

Stock/Pending count Full accounts with residual shares/receivables; zero Full residuals do not imply none for other strategies. All 6,120 account executions are in the companion results. Fee/slippage, advance fees, sell charge, lots and capacity are assumptions. No-fill/delay proxies do not measure queue fills; all entries missed is not a mathematical worst bound.

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